As-Continuous-As-Possible Multi-Material Extrusion with Gradient Composition and Transition

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Abstract

Gradient Multi-material extrusion enables the efficient production of locally tailored materials. On large scales for construction purposes, relevant studies are limited in shape complexity, material count, and design methods. On the sub-millimeter scale of fused filament fabrication, current methods do not mix filaments to create gradient materials, creating seam defects and considerable material waste. This study is an effort to connect the two paradigms under one numerical model on the material mixing and transitional behavior in gradient multi-material extrusion, focusing on the multi-material toolpath computation methods. Built on single-nozzle multi-filament extruding systems and material-coded toolpath processing algorithms, this study proposes the method of gradient composition and transition to efficiently produce 3D objects of complex materials and shapes across scales. By incorporating an as-continuous-as-possible planning algorithm, the method is capable of generating 3D multi-material toolpaths with minimal seam defects and material wastes. The methods for generating 2D continuous space-filling multi-material toolpaths with soft- or hard-edge effects are also illustrated. Finally, the proposed method is implemented as a plug-in called Ovenbird in Grasshopper® for Rhino®, which is compatible with different design inputs and streamlines multi-material 3D printing tasks for designers and engineers.

Keywords*: Multi-material extrusion, 3D printing, fused filament fabrication, toolpath planning, design automation*

1. Introduction

1.1. Large-Scale Multi-Material Extrusion

Large-scale multi-material extrusion is a promising technology for the construction industry, allowing structures or structural components to be efficiently produced with locally tailored materials. Several studies explored single-nozzle gradient multi-material extrusion systems that control the composition of concrete and reinforcement materials (fiber or aggregates) to realize functionally graded materials for structural optimization and embedded functionalities (Pajonk et al. 2022; Ahmed et al. 2022; Craveiro et al. 2020; Duballet et al. 2015). However, limitations and open questions remain in these studies:

- To avoid visually prominent seam defects (Yigit and Lazoglu 2019), researchers have opted for simple cylindrical geometries without stop-starts, thereby limiting the geometrical form of the print, thus neutralizing one of the key benefits of 3D printing (Bos et al. 2016). Strategies to make toolpaths as continuous as possible have been extensively explored in large-format single-material extrusion (Hergel et al. 2019; Zhong et al. 2023) but are yet to be transferred to the multi-material side.
- Only two material inputs (or "filaments") are used to form a monotone linear spectrum of material composition. The method to effectively encode and deliver three or more filaments for broader applications has not been explored.
- The methods to determine and encode the local material compositions are limited. In the case studies, the material compositions are determined by simple boxes, 1D fields, or voxel grids of low resolution. Alternative methods that are visually more intuitive will contribute to the precise control of local material composition and thus assist designers and engineers in design iterations.

1.2. Colored Fused Filament Fabrication

On the other hand, researchers and engineers of fused filament fabrication (FFF) printers have developed software and hardware that produce multi-filament colored print at millimeter- to centimeter-scale. The commercially successful multi-filament printers, coupled with slicing software of polished user interfaces, have allowed hobbyists to model and print colored parts with high-quality (Song et al. 2019). To connect such success to broader applications at the construction scale, we highlight the following issues:

- While these printers can switch between 8 or more filaments during the print (Anycubic 2024; Bambu Lab 2024a), they are not capable of mixing multiple filaments to create *gradient* colors, nor does the accompanying design software.
- As thermoplastic filaments are typically extruded at a sub-millimeter section, the seam defects are unnoticeable and toolpaths are not generated to be as continuous as possible.
- To ensure accurate deposition of distinct filaments, the extruder uses a wipe tower for priming when switching between filaments (Bambu Lab 2024b; Prusa 2024). In large-scale material extrusion, this would introduce a significant waste of materials.

1.3. The Gradient Composition and Transition Method

This study stands at the intersection of the two aforementioned paradigms of multi-material extrusion. Figure 1 summarizes the multi-material extrusion research landscape in terms of material/color and shape complexities found in both paradigms. The ultimate goal of (functionally) gradient multi-material extrusion is to fabricate "full color" objects of complex shapes with precise per-"voxel" material/color deposition, similar to the effect of 2D/3D inkjet printing. This necessitates clean, sharp transitions between material gradients where the bonding strength is preserved. Since the length of the transition is not negligible, the desired image/scheme can only be approximated. This study proposes *the gradient composition and transition method* to achieve the approximation. It (i) mixes the filaments to create a discrete, gradient palette and (ii) assigns the palette to an aggregation of monolithic regions between which transitions take place. In addition,

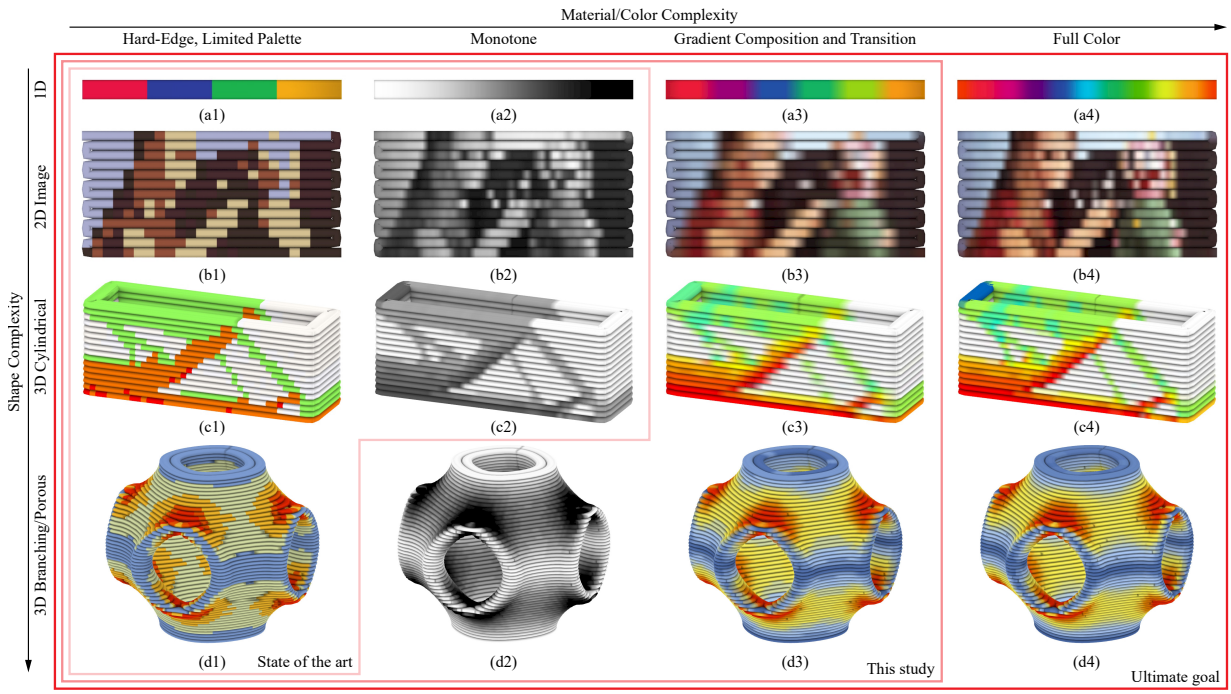

Figure 1: The landscape of multi-material extrusion research. The extruded profiles of the examples are exaggerated to show the trace of 3D extrusion. (a) 1D rainbow gradient; (b) 2D image, part of the BBC test card featuring Carole Hersee (owned by George Hersee, used under fair use for educational and informational purposes); (c) 3D cylindrical geometry, an optimized half-MBB (Messerschmitt-Bölkow-Blohm) beam with colors showing the in-plane principal stress as calculated by ABAQUS (Heng et al. 2024); and (d) 3D branching/porous geometry, one cell of Schwartz P triply periodic minimal surface (TPMS) modeled using Axolotl (Bernhard et al. 2018), with colors showing the local overhang (buildability) (Zhi et al. 2025).

by introducing *as-continuous-as-possible* gradient multi-material extrusion, this study fabricates objects that combine gradient composition and transition with complex 3D shapes (Figure 1) while minimizing seam defects and material waste. In the following sections, this study will:

- Provide a numerical model on the material mixing and transitional behavior in gradient multi-material extrusion and achieve precise control over the material composition of 3 or more filaments using a single nozzle (Sec. Section 2).
- Offer a ready-to-use modeling interface that collects and organizes material information from different sources, generates multi-material toolpath and GCodes, and offers visualization tools (Sec. Section 3).
- Minimize the waste in 3D multi-material extrusion by combining a planning algorithm for as-continuous-as-possible MMTPs with an alternative wipe unit design (Sec. Section 4).

- Create two types of 2D continuous space-filling MMTPs that embody a colored image with soft-edge, motion-blur-like effects or hard-edge posterization effects (Sec. Section 5).

By fulfilling these objectives, this study introduces gradient printing to fused filament fabrication and opens up new possibilities for large-scale functionally graded/gradient multi-material extrusion.

2. Gradient Multi-Material Extrusion

2.1. Hardware and Materials

2.1.1. Extruding System

The proposed gradient multi-material extrusion relies on a single-nozzle multi-filament (SNMF) system where a palette of input filaments or pastes are fed and mixed to form the extrudate with tailored local material composition (Figure 2). The core of the SNMF system is a multi-inlet nozzle where filaments/pastes are fed through separate channels and converged at the mixing chamber equipped with a blending auger.

A customized firmware based on Marlin 2.1.2 drives the stepper motors, regulating the relative feeding rate of individual filaments for precise material composition control. Integrated heating elements and thermistors ensure uniform material melting when printing with thermoplastics.

2.1.2. Material and Scalability

While the case studies in this paper are prepared using the SNMF system of thermoplastic filaments (Figure 2a), mainly polylactic acid (PLA), the proposed method also applies to paste materials (Figure 2b) where the filament rolls are replaced by syringes and hoppers (Teng et al. 2024). Each input material, instead of being driven by the filament supplying motor (E1 in Figure 2a), is driven by both the plunger (E3 in Figure 2b) and the feeding auger (E1 in Figure 2b) in the extruding head. Successful extrusion and bonding usually require similar viscosity in the pastes, such as clay and

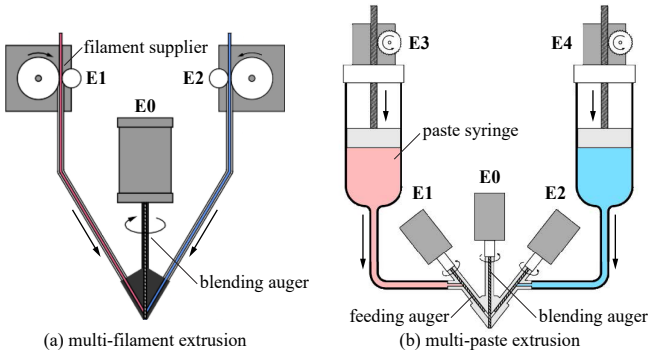

Figure 2: Schematic drawing of gradient multi-material extrusion systems with active mixing (E0) for (a) filament and (b) paste. Adapted from Teng et al. (2024) and Teng et al. (2025)

diatomaceous earth. The successful multi-material extrusion of dissimilar paste materials remains an open field of study.

2.2. Embedding Gradient Mixtures

A mixture of n filaments can be represented by a stochastic row vector $\boldsymbol{v} \in [0,1]^n$ with the components specifying the fraction of each filament. The desired composition of the mixture is a local property. Naturally, $\boldsymbol{v}$ is a continuous function over the nozzle travel length x.

To efficiently represent and store the desired mixtures, we (i) collect and define the vectors all at once to form the discrete, gradient palette; (ii) insert vertices in the toolpath polygons at desired positions of material change; and (iii) specify the index of the new mixture at those vertices. In the first step, when we can have m mixtures over n filaments, the mixtures are stored in the in-house slicing software as an $m \times n$ matrix

$$\boldsymbol{P} = \begin{bmatrix} \boldsymbol{v}_1 \\ \boldsymbol{v}_2 \\ \vdots \\ \boldsymbol{v}_m \end{bmatrix}. \tag{1}$$

In the multi-material GCode implementation based on the Marlin 2.1.2 firmware (Teng et al. 2023; Teng et al. 2025), each unique mixture is stored as a "tool" and called when the nozzle travels to a position using it. This allows efficient memory management since tools form a palette and the same "tool" will be called repeatedly in extruding complex objects.

2.3. Numerical Model of Gradient Transitions

2.3.1. Filament and Mixture

To evaluate gradient transitions, consider another stochastic row vector $\hat{\boldsymbol{u}} \in [0,1]^m$ that specify the fractions of the m mixtures in Eq. (1). The resulting composition of the n filaments is given by the stochastic vector

$$\boldsymbol{u} = \hat{\boldsymbol{u}}\boldsymbol{P} \in [0,1]^n. \tag{2}$$

Recall the nozzle travel length x. In this manner we define (i) the design input $\hat{\boldsymbol{d}}(x)$ and $\boldsymbol{d}(x)$; (ii) the relative supplier motor speed $\widehat{\boldsymbol{m}}(x)$ and $\boldsymbol{m}(x)$; and (iii) the resulting composition $\hat{\boldsymbol{r}}(x)$ and $\boldsymbol{r}(x)$. Naturally $\hat{\boldsymbol{d}}(x)$ has only one non-zero component specifying the index of the mixture used.

2.3.2. Motor Speed and Resulting Mixture/Filament Composition

To allow gradient composition and enhance strength at material boundaries, our hardware design incorporates an active mixing nozzle. It rotates at a constant speed during the print and introduces a delay in the gradient material transition between mixtures. A previous study (Teng et al. 2025) shows that the resulting gradient transition can be characterized as a moving average with lag

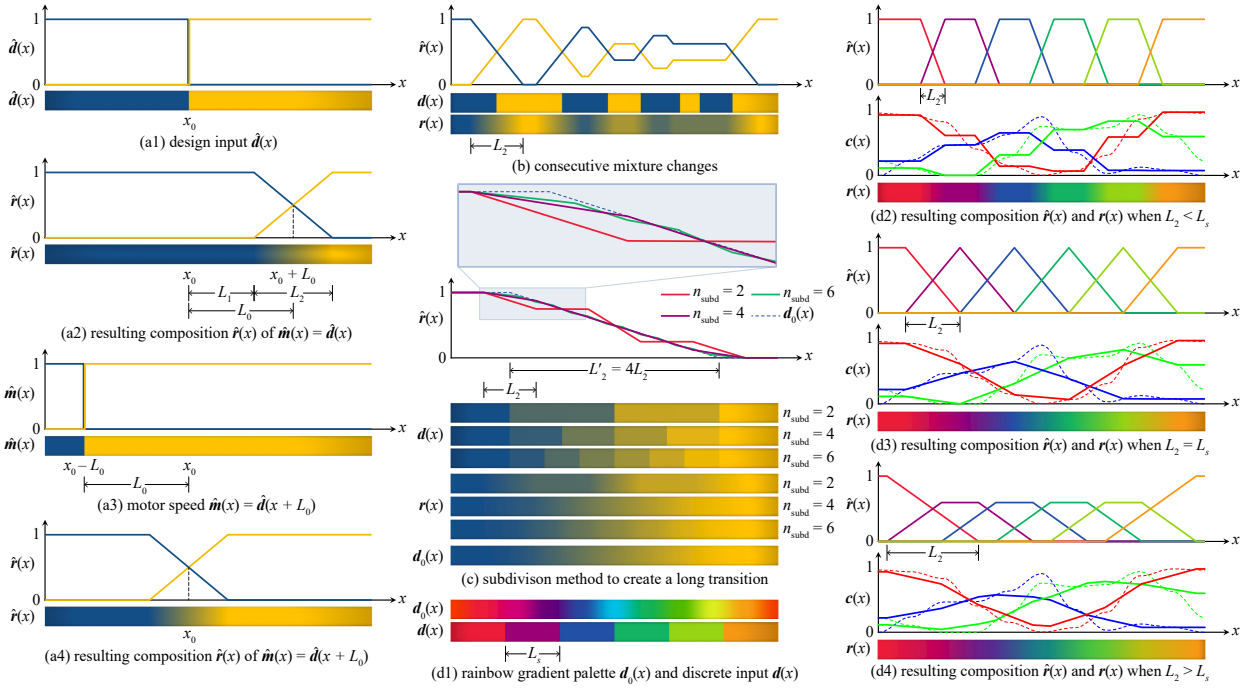

Figure 3: Material transition studies. (a) The relation between motor speed and resulting mixture composition, showing the fractions of two mixtures (components of the vector). (b) Result of consecutive mixture changes. (c) The subdivison method to create a long transition of length $L_2' = 4L_2$. The line graph shows only the fraction of the first material. (d) Resulting mixture composition and mixture colors (RGB channels) in gradient transitions between $m = n = 6$ source colors.

$$\hat{\boldsymbol{r}}(x) = \frac{1}{L_2} \int_{x-L_1-L_2}^{x-L_1} \widehat{\boldsymbol{m}}(x)\, \mathrm{d}x \tag{3}$$

where L_1 and L_2 are the *delay length* and the *transitional length* which depend on the nozzle design and the printing speed (Figure 3a1 and a2). They can be measured in printing tests adopting the natural strategy that the speed of the material feeding motors follows the target mixture compositions (Teng et al. 2025)

$$\widehat{\boldsymbol{m}}(x) = \hat{\boldsymbol{d}}(x). \tag{4}$$

2.3.3. Transition Adjustment

We want to compensate for the lag introduced in Eq. (3) to match the print result with our design. Consider the case of transitioning at breakpoint $x = x_0$ (Figure 3a1):

$$\hat{\boldsymbol{d}}(x) = \begin{cases} [1\ \ 0] & x < x_0, \\ [0\ \ 1] & x \geq x_0. \end{cases} \tag{5}$$

From Eqs. (4) and (5), the midpoint of the resulting transition is at $x = x_0 + L_1 + \frac{1}{2}L_2$ where $\hat{\boldsymbol{r}}(x) = [0.5 \;\; 0.5]$ (Figure 3a2). Thus we may move backward the motor breakpoint by *advancement length* $L_0 = L_1 + \frac{1}{2}L_2$ so that the resulting/visual breakpoint aligns with the design input (Figure 3a3 and a4):

$$\widehat{\boldsymbol{m}}(x) = \hat{\boldsymbol{d}}(x + L_0). \tag{6}$$

2.3.4. Discrete Form

In the proposed data sturcture of $\widehat{\boldsymbol{m}}(x)$, mixture changes are assigned only at specific locations $x_1 < x_2 < ... < x_p$. If $x_i \le x < x_i + 1$, $\widehat{\boldsymbol{m}}(x) = \widehat{\boldsymbol{m}}(x_i)$. Let $\ell(x)$ be the largest index i such that $x_i \le x$, then $\widehat{\boldsymbol{m}}(x) = \widehat{\boldsymbol{m}}(x_{\ell(x)})$. Following this, the pratical, discrete form that calculates the resulting composition as in Eq. (3) is

$$\hat{\boldsymbol{r}}(x) = \sum_{i=\ell(x-L_1-L_2)}^{\ell(x-L_1)} f(i,x)\, \widehat{\boldsymbol{m}}(x_i) \tag{7}$$

where $f(i,x) \in [0,1]$ is the contribution of the ith mixture change to the location x:

$$f(i,x) = \frac{\min\{x_{i+1}, x - L_1\} - \max\{x_i, x - L_1 - L_2\}}{L_2}. \tag{8}$$

2.4. Simulation and Visualization

This section discusses the practical way to simulate and visually represent the result of gradient transitions using the proposed model Eq. (7). Consider two points $x_a < x_b$. If $\ell(x_a - L_1 - L_2) = \ell(x_b - L_1 - L_2)$ and $\ell(x_a - L_1) = \ell(x_b - L_1)$ then the resulting composition at any point between the two can be solved through linear interpolation. Thus the line graphs of the components of $\hat{\boldsymbol{r}}(x)$ and $\boldsymbol{r}(x)$ are in polygon shapes. The vertices are at

$$x_i + L_1 \quad \text{and} \quad x_i + L_1 + L_2, \qquad i = 1, 2, ..., p. \tag{9}$$

These vertices are added to the toolpath curve for visualization, although they might not be turning points. The color at all toolpath vertices can be then solved using Eq. (7). This takes linear time since the bounds of index i increases monotonically. Then we can visualize the color at all points using the method of mesh vertex coloring. Single-material extrusions can be efficiently represented as meshes where each toolpath vertex corresponds to a planar polygon diaphragm (Zhi et al. 2025). The corresponding mesh vertices can be colored per section and the faces can be rendered gradiently.

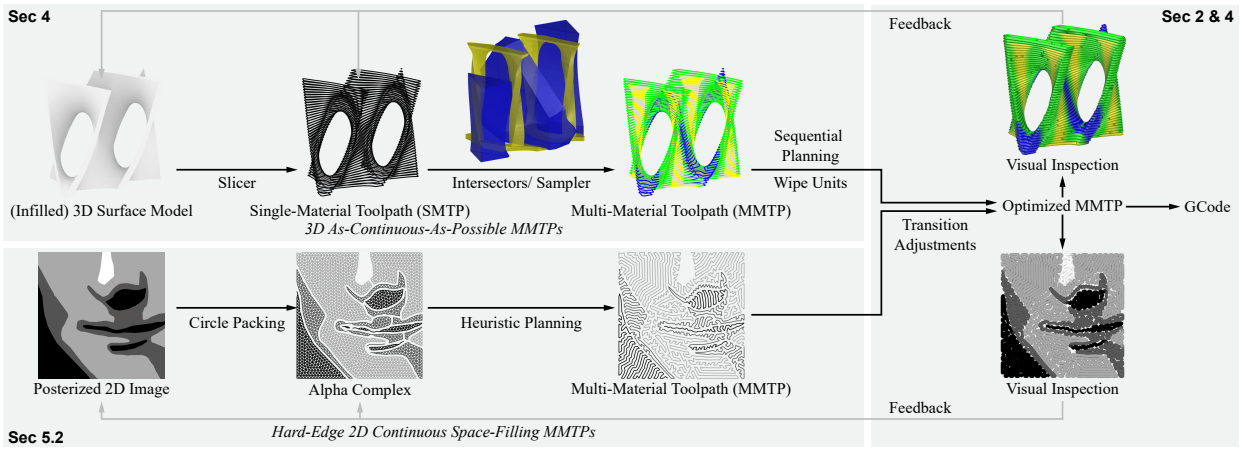

Figure 4: Computation workflows of 3D as-continuous-as-possible MMTPs (Sec. Section 4) and hard-edge 2D continuous space-filling MMTPs (Sec. Section 5.2). The 3D surface model is a scaled brick of the Diamond TPMS with the color showing the stress if the structure is under compression. The 2D image is part of the *Mona Lisa* by Leonardo da Vinci. 3D and 2D MMTPs share the same 3D multi-material data structure organized in layers, with the 2D case having only one layer.

2.5. Compound Effects of Gradient Transitions

2.5.1. Consecutive Mixture Change

Figure 3b shows the simulation of rapid, consecutive changes between two mixtures with gaps smaller than the transitional length L_2. The extrusion might not reach 100% of the desired mixture before it turns to another mixture. Some designated locations of material change have been completely blurred out after the moving average filter is applied.

2.5.2. Extended Transition

It is observed that for the stability of extrusion, the blending auger should have a fixed rotational speed. Therefore the transitional length L_2 is a fixed value. Consider the task of creating a very long gradient transition of length $L_2' \gg L_2$ between two mixtures, which can be found in functional graded objects. Figure 3c illustrates an example where $L_2' = 4L_2$. The transition region can be subdivided into n_{subd} blocks; each assigned an intermediate mixture between the two parent mixtures. Both the line graph and the visualization show that when $n_{\text{subd}} \approx L_2'/L_2$ the subdivision method offers the closest approximation.

2.5.3. Color Blending in Transitions

To speculate the effect of mixing different colors in transitions, a rainbow gradient palette is used as a benchmark (Figure 3d1). A discrete palette of $m = n = 6$ colors is sampled, each covering a

desired print length of L_s. The RGB channels of the desired palette and the gradient composition and transition results are calculated. Figure 3b—d compares the effect of different transitional length L_2 in blending the 6 colors and recreating the continuous spectrum. When $L_2 \ll L_s$, the segments appear as chunks. When $L_2 \gg L_s$, the colors are over-blended and result in an off, deep spectrum. For full-color printing, the resolution is thus limited by L_2.

3. Multi-Material Toolpath (MMTP) Computation Overview

Two major approaches exist in computing multi-material toolpath (MMTP): (i) Assigning materials on a single-material toolpath (SMTP) to form an MMTP and (ii) Generating SMTPs in single-material regions and connecting them to form an MMTP. Figure 4 outlines two proposed MMTP computation workflows that fall into these two categories, showing the input and output data, methods used, and feedback loops offered by the workflow.

The proposed workflow of generating as-continuous-as-possible gradient MMTP is implemented as a plug-in software called Ovenbird (Zhi 2024) (Figure 5) in the modeling and visual programming environment Grasshopper® for Rhino®. It offers an agile, code-free user interface for designers to prepare MMTP while seamlessly harnessing their advantages in modeling and managing complex geometries. With the proposed system and software, gradient multi-material objects can be designed and tested out for purposes such as colored fabrication, structural reinforcement (Figure 1c), and scientific communication (Figure 1d).

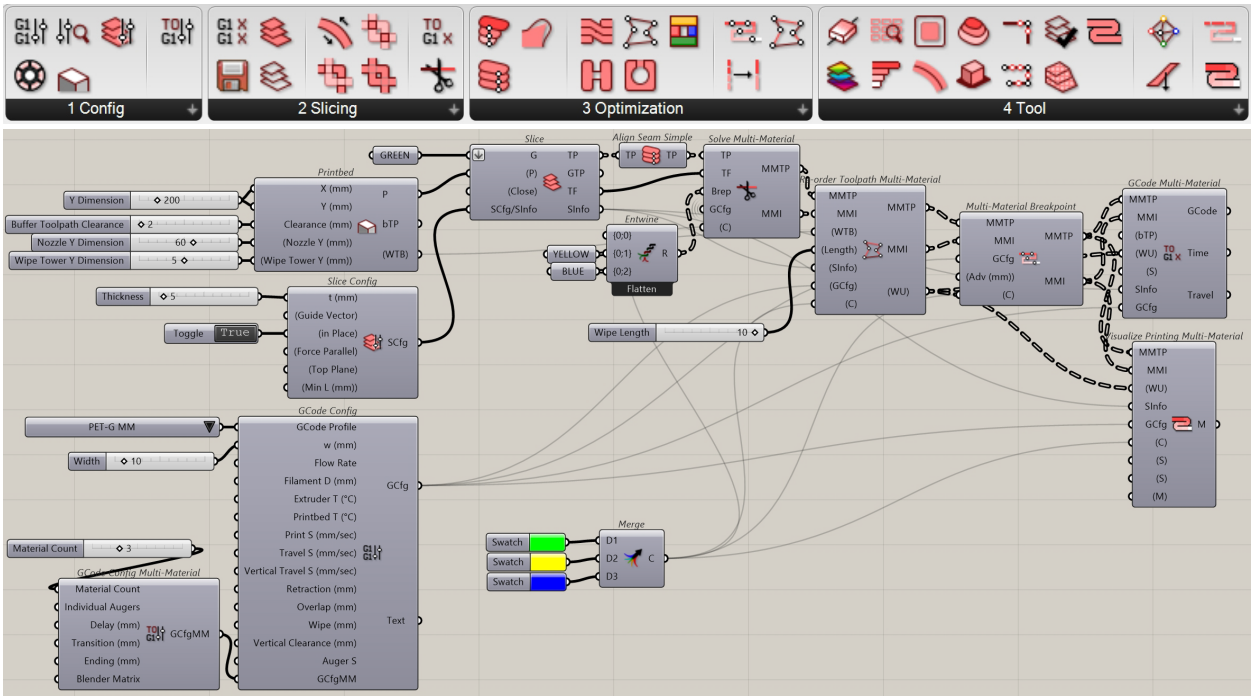

Figure 5: Top: Component panel of the plugin for Grasshopper®. Bottom: User interface of a minimal working example that generates the 3D MMTP in Figure 4.

4. 3D As-Continuous-As-Possible MMTPs

3D SMTP typically uses a hierarchical data structure organized in layers (Zhi et al. 2023; 2025). This section introduces the MMTP data structure and the method of assigning materials on an SMTP to form an MMTP.

4.1. Assigning Materials

4.1.1. 3D Multi-Material Data Structure

In single-material extrusion, each toolpath curve is extruded continuously, following the winding direction from the start to the end. Each layer may consist of multiple curves. In the SMTP data structure, each curve is indexed as $\{\text{layer}, \text{curve}\}$ (Figure 6a). In multi-material extrusion, a curve may consist of multiple segments assigned with distinct materials (in gradient multi-material extrusion, mixtures). In the MMTP data structure, each segment is indexed as $\{\text{layer}, \text{curve}, \text{segment}\}$ and marked with a material (Figure 6b). The curves of SMTP in the given example are broken into single-material segments connected tip to tail. Depending on the source of material information, two methods of assigning materials can be used.

4.1.2. The Intersector Method

When distinct materials are desired for some explicit regions, it is handy to model boundary-represented solid objects (mesh, NURBS, etc.) as intersectors and intersect them with the SMTP to break them up (Figure 6c). A segment is assigned a specific material if it is inside a corresponding intersector. For a m-material palette, $m-1$ sets of intersectors are required. For instance, a truss may prefer regular materials at compressive regions and reinforced materials at tensile regions.

4.1.3. The Sampler Method

When materials are expected to have a wide range of gradients or the material information is given implicitly, the sampler method can be used. The SMTP is divided into small segments of a certain resolution. The division can follow voxel grids or use rules such as maximal segment length. The material information is accessed by sampling per segment. When the material is given as a continuous spectrum while a discrete palette is used for extruding, the sampled material can be mapped to the closest preselected mixture. After the MMTP is formed, consecutive segments without material changes can be merged for simplicity. In Figure 6, both (c) and (d) result in (b). In the intersector method, the accurate modeling of the intersector can be challenging while in the sampler method, the result heavily depends on the resolution.

4.2. Sequential Planning

4.2.1. Multi-Material Seam Defect

For multi-material closed curves, the original seam in SMTP is naturally replaced by one of the material changes in MMTP to minimize the number of single-material segments (Figure 6). Consequently, the curve starts and ends with different materials, which introduces a visible multi-

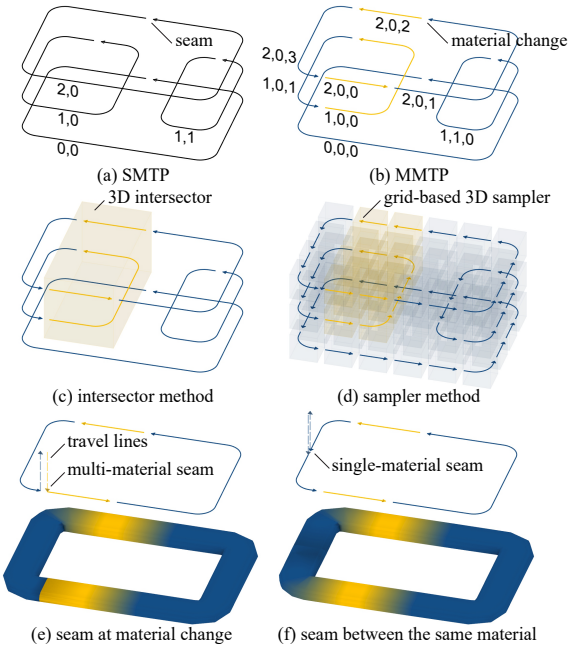

Figure 6: (a)–(d) A 3-layer minimal example showing the data structures and material assignment methods. Gaps between layers are exaggerated. (e), (f) Multi-material seam defect. Space between segments is added for display.

material seam defect without gradient transition, visualized in Figure 6e and f. This defect can be replaced with gradient transition by having the seam of the multi-material curve at the middle of any of the single-material segments. Now the nozzle will enter and exit with the same material, called the *transit material*.

4.2.2. Wipe unit

When the nozzle travels between two curves of different transit materials, a wiping stop is needed. The nozzle typically extrudes somewhere off the print body to complete the material transition. For each transition, we place a rectangle curve known as a *wipe unit* (Figure 7) whose length should be greater than the transitional length L_2, preferably with a factor.

4.2.3. Sequential Planning Problem

Cross-layer continuous printing of branching, porous geometry requires a slender, long nozzle (Zhi et al. 2025), which is incompatible with multi-material filament extrusion as the mixing chamber must stay close to the outlet to prevent clogging. Sequential planning thus aims to minimize multi-material seam defects and the number of accompanying wipings. (While in double-sided/infilled situations wipings can be placed in hidden parts, it is computationally unstable and conflicts with the goal of functionally graded material. We thus prefer stand-alone wiping units.) We also adopt the assumption that all curves are closed. Consider the following formulation of the problem:

Input: A universe of m materials and an MMTP consisting of several layers. Each layer consists of several curves. Each curve covers a set of materials, all candidates of the curve's transit material. *Goal:* An assignment of transit materials over curves and the ordering of curves in each layer such that the number of wiping stops is minimized for the entire print.

Whenever the nozzle carries a specific material, it remains optimal if the nozzle greedily goes through all curves covering that material before future planning. Following this observation, if we only have one layer, the problem is equivalent to the famous hitting set problem, which is NP-complete (Korte and Vygen 2012). Hence, we adopt a best-first greedy algorithm for the global problem. Upon deciding the next transit material, the algorithm traverses all materials and picks the one that allows most layers and curves to be printed.

After the transit material of a curve is decided, the longest segment of that material is divided for the nozzle to enter and exit. This minimizes unwanted blending and contamination.

4.3. Minimal Wiping

While the greedy algorithm does not guarantee global optimal, the number of wiping stops needed is usually tiny and significantly less than the number of layers in producing functionally graded objects. In a conventional off-the-shelf multi-material extrusion system, the size of the wipe tower is decided by the maximal number of wipes within a layer. This introduces significant waste as the material has to be extruded as support. Our approach uses wiping units only if needed and is compatible with open bed systems such as gantry and robotic arm. Figure 7 illustrates the proposed method where the print zone and the wipe zone are separated by a band parallel to the beam, along the x axis. The width of the band d_1 is greater than the distance between the nozzle tip and the

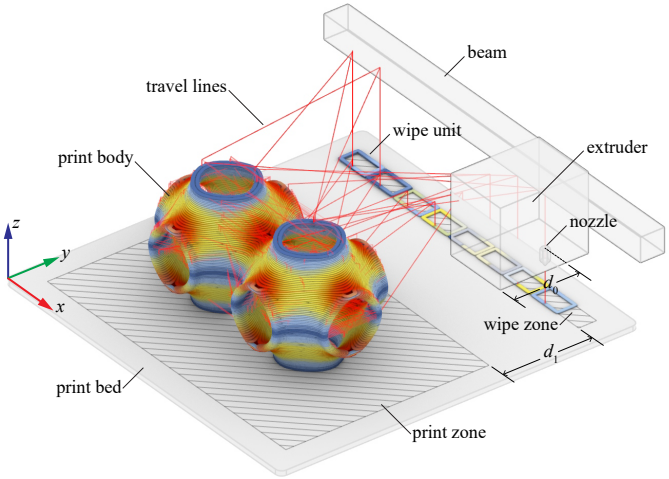

Figure 7: Print scheme of two cells of Schwartz P TPMS on a 300×300 mm print bed, showing the wipe units. This 90-layer print uses 8 wipe units. Compared with a traditional wiping tower (at least 1 unit per layer), it saves $1 - 8/90 = 91\%$ of the wiping material, 11% of the total material. Timewise, it still saves 13.8% of the traveling time despite the large clearance added between the print and wipe zones.

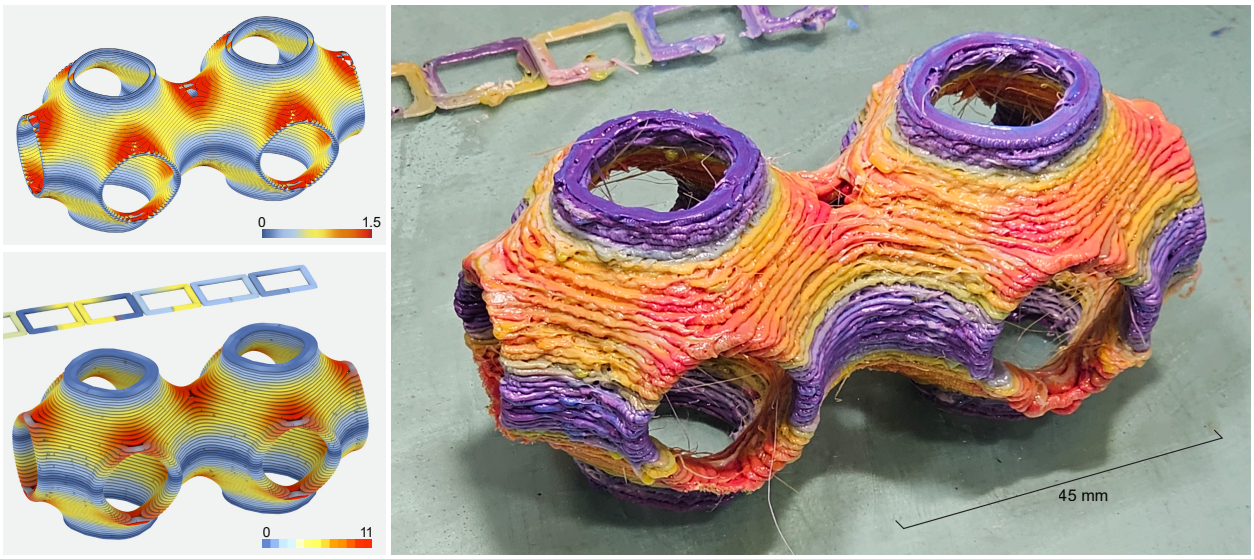

Figure 8: 3D gradient multi-material extrusion of 45-layer colored Schwartz P TPMS cells. Left top: Sampler showing the print's local overhang (Zhi et al. 2025). Left bottom: Visualization of the print body, wipe units, and the palette used ($m = 12$). Process of printing. Right: The finished print and wipe units on the print bed.

bounding box of the extruder d_0 so that the extruder goes down at the wipe zone without colliding existing print body. The wipe units do not have to be stacked up to match the height of the current print body during the print. No modification to the motion system is needed.

Finally, before converting the MMTP into GCodes, the transition adjustment discussed in Sec. Section 2.3 takes place for accurate material deposition. Each breakpoint introduces a blurred-out zone of length L_2. When the nozzle travels between two curves of the same transit material, if either the last segment of the first curve or the first segment of the second curve is shorter than $L_2/2$, a third material will be blended into the other curve. Therefore, an extra wipe unit is added. The print result (Figure 8) shows that the wipe units collect all unwanted material changes. The main body has a consistent coloring. The overhang-based coloring makes it a tool for design communication and education.

5. 2D Continuous Space-Filling MMTPs

2D continuous space-filling MMTPs that recreate a desired image input can be generated through both approaches stated in Sec. Section 3, achieving different effects (Figure 9).

5.1. Soft-Edge Parallel Filling

Zigzag parallel fillings are widely used in additive manufacturing as they are easy to compute and have consistent tool orientation. Through either the intersector method or the sampler method, just as in 3D, material information can be added to zigzag parallel fillings (Sec. Section 4.1). Transitions between materials take place in parallel directions, creating a motion blur effect (Figure 9). In 3D

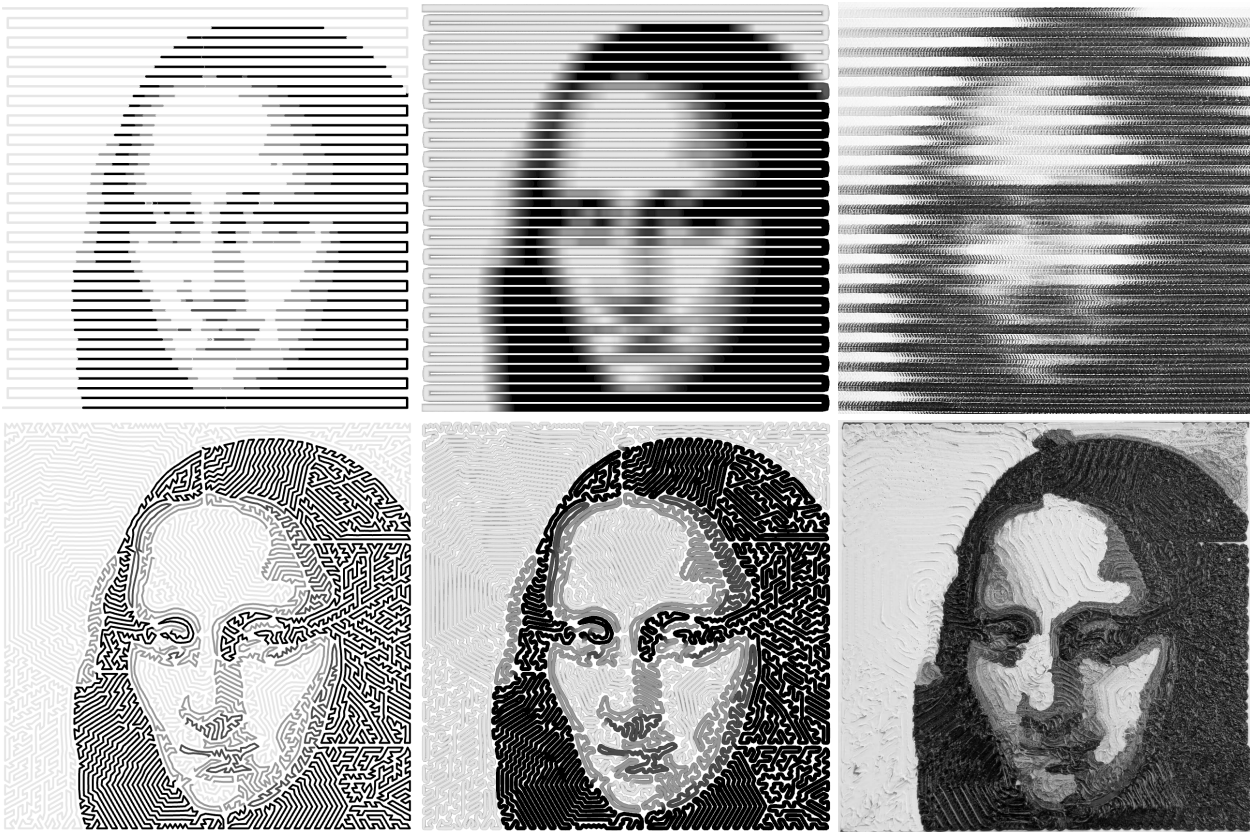

Figure 9: 200×200 mm **2D gradient multi-material extrusion of a posterized image from the *Mona Lisa*. Top row: Soft-edge parallel filling. Bottom row: Hard-edge alpha complex filling. Left: Toolpath before transition adjustments. Middle: Visualization. Right: The finished prints.**

manifolds, curves of adjacent layers can have their winding directions unified (Zhi et al. 2025) to make the error in transition adjustment uniform. In 2D zigzag parallel filling curves, the adjacent segments have opposite directions, making the error highly visible (Fig Figure 9). A large number of material transitions is also not ideal in circumstances when the gradient material transition is subject to delamination.

5.2. Hard-Edge Alpha Complex Filling

The alternative method of hard-edge filling (outlined in Figure 4) uses minimal transitions, creating minimal defects at material transitions (Figure 9). It requires that, for an image made of single-material regions, each region is filled independently with a single curve. Methods such as connected Fermat spirals (Zhao et al. 2016) and globally continuous hybrid path (Xia et al. 2022) fulfill this task for a single region. We introduce the *alpha complex filling* method that deals with intricate, small single-material regions and creates globally continuous space-filling multi-material toolpaths.

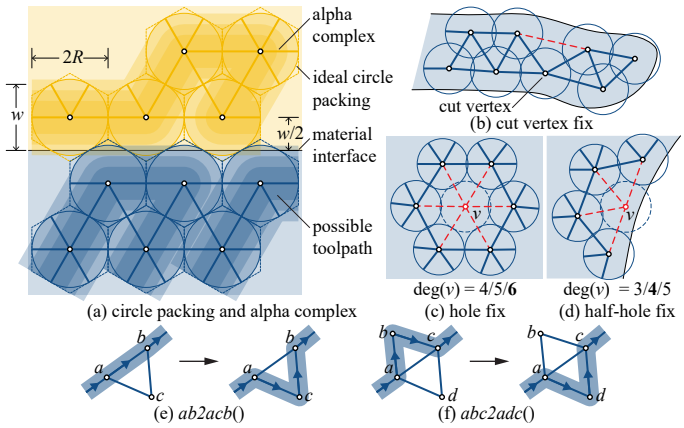

Figure 10: (a)–(d) The generation of alpha complex for single-material regions. (e), (f) Local relaxations for finding a Hamiltonian path on an alpha complex.

5.2.1. Alpha Complex

An alpha complex is a 2D graph data structure built between vertices that reflects their proximity. For a hard-edge image, regions and their boundary curves can be extracted using tools such as contour solvers (OpenCV 2025). The alpha complex "fills up" a region like a triangular mesh if the vertices fill up the region evenly, which can be the result of circle packing. Then if the nozzle travels along a Hamiltonian path of the alpha complex, the extrusion will fill up the region. Let w be the extrusion width and R the radius of the circles. Ideally, each vertex covers a hexagon area of $A_h = 2\sqrt{3}R^2$ and the nozzle travels along a $2R$ long segment inside the hexagon (Figure 10a). For the extrusion to fill up the area, we have $2Rw = A_h$, which gives $R = w/\sqrt{3}$.

5.2.2. Circle Packing

We adopt a non-strict circle packing model where circles are allowed to overlap a little bit. This can be achieved by physical simulation. In a single-material region of area A_r, A_r/A_h circles are populated randomly. Then a collision model allows the vertices to have a $2R$ gap between each other and a $w/2$ gap between the material interface (Figure 10a). In our implementation, the physical simulation uses Kangaroo on Grasshopper® (Piker 2017). After the model converges, we can solve the Delaunay triangulation and cull the long edges to form the alpha complex.

The result might have two types of defects. Cut vertices might exist, which require the path to start and end on two sides of the region (Figure 10b). Holes (Figure 10c) and half-holes (Figure 10d) might be left, making the region underfilled. Both defects can be located in linear time since the graph is planar and the maximal degree is 6. They can be fixed by adding edges and vertices.

5.2.3. Global Hamiltonian Path

To form the global Hamiltonian path, the Hamiltonian path connecting all regions is solved first. When the number of regions is large, it can be solved heuristically. If the graph between regions

is not Hamiltonian, the regions can be manually subdivided. When the number of regions is small, it can be solved using brute force algorithms. Extended research on this task is seen in (Zhai and Chen 2019). Based on the global path, locations of entering and exiting each region can be decided. A primitive path connecting the two is given by depth-first search (DFS). By iteratively relaxing it (Figure 10e and f) in a simulated annealing manner, the space-filling path can be solved. Referring to the theorem that every 4-connected planar graph has a Hamiltonian path between any two vertices (Thomassen 1983) and that our alpha complex is "well" connected, this task is very likely to succeed.

6. Conclusion and Future Work

This exploratory study introduces the gradient composition and transition method of multi-material extrusion. The method of creating as-continuous-as-possible gradient MMTPs with minimal waste and defect has been tested. The software for efficient design and visualization is developed. While the examples are all fabricated using thermoplastics and FFF printers, the method can also be tested using paste extrusion systems.

We highlight the following research interests for future work. The hardware can be upgraded for precise deposition and improved extrusion quality. Alternative numerical models on gradient composition and transition can be explored and compared with the proposed one. Schemes for accurate color mixing can be studied for both filament and paste as in (Song et al. 2019). A comprehensive review of the design strategies and applications of gradient multi-material objects produced by our method is also needed.

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